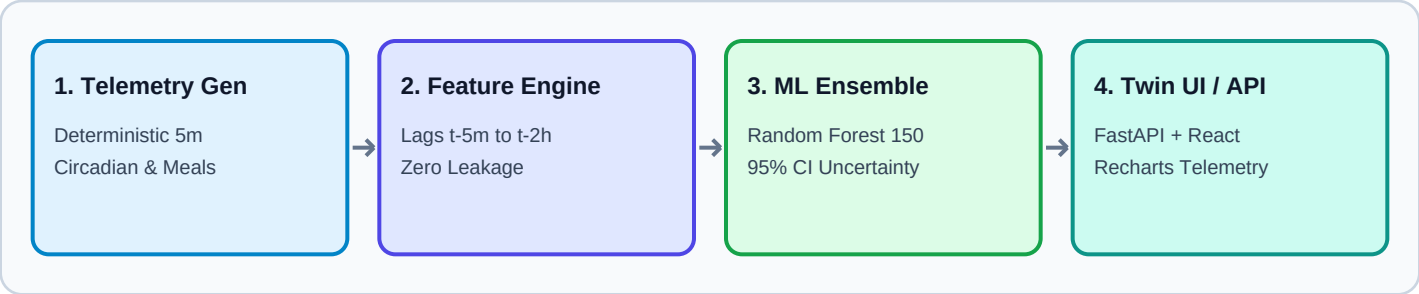
GlucoTwin AI — Architecture Blueprint

A Time-Series Digital Twin for 2-Hour Glucose Trend Forecasting | Unstop Digital Twin Challenge 2026

CRITICAL SAFETY & SCOPE NOTICE: *This system is an educational prototype and research demonstration. Not a medical device. Never for clinical diagnosis, treatment decisions, glycemic emergencies, or insulin dose titration. All data is synthetically generated.*

1. Executive Overview & Research Need

Continuous Glucose Monitoring (CGM) enables dense time-series observation of metabolic states. However, raw reactive monitoring provides limited forward visibility. GlucoTwin AI establishes an in-memory virtual patient digital twin that integrates historical glycemic context, meal carbohydrate loads, and physical exertion to forecast glucose trends **2 hours (120 minutes) into the future** with calibrated 95% confidence intervals.



2. Four-Tier Pipeline Specifications

Tier 1: Synthetic Telemetry Engine — Generates continuous 5-minute sampling observations incorporating meal absorption delays (exponential convolution), physical exertion uptake, heart rate scaling, and circadian baseline oscillations.
Tier 2: Feature Engineering & Leakage Prevention — Computes 23 strictly backward-looking time-series features: lags (t-5m, t-10m, t-15m, t-30m, t-1h, t-2h), rolling averages, derivative velocity (15m/30m rates of change), and harmonic sin/cos representations of time.
Tier 3: Machine Learning Ensemble — Trains a 150-tree RandomForestRegressor with chronological 80/20 train/test split. Quantifies uncertainty from tree variance: standard error yields mathematically defensible 95% confidence intervals.
Tier 4: FastAPI REST API & React Digital Twin — Asynchronous REST services exposing state, history, 120-minute predictions, and What-If simulation sandbox backed by a modern React + TypeScript + Recharts dashboard.

3. Empirical Evaluation vs Persistence Baseline

Metric	Trained ML Model	Persistence Baseline (y(t+24)=y(t))	Honest Improvement (%)
Mean Absolute Error (MAE)	5.260 mg/dL	48.711 mg/dL	+89.20% Error Reduction
Root Mean Squared Error (RMSE)	7.160 mg/dL	59.482 mg/dL	+87.96% Error Reduction
Holdout Evaluation Samples	1,091 observations	1,091 observations	Chronological (Zero Leakage)

4. What-If Simulation Sandbox & Clinical Translation Roadmap

The What-If module projects hypothetical perturbations (e.g. 45g carbohydrate intake alongside 60% physical exertion) over the 2-hour window. It provides immediate visual and quantitative feedback while explicitly labeling outputs as exploratory.

Before clinical translation, this digital twin requires: (1) validation on clinical trial datasets (e.g., OhioT1DM or D1NAMO), (2) sensor lag deconvolution for interstitial-to-blood glucose delay, and (3) formal Software as a Medical Device (SaMD) regulatory evaluation.